Answer Key - Physics - Undergraduate Level Assessment 14 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) 0.50 c; B) 0.60 c; C) 0.70 c; D) 0.80 c
   Note: Correct option: B - 0.60 c

2. **Answer:** B
   Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic
   Note: Correct option: B - Constant volume

3. **Answer:** D
   Options: A) square; B) hexagon; C) cube; D) tetrahedron
   Note: Correct option: D - tetrahedron

4. **Answer:** D
   Options: A) 0.1 GeV/c^2; B) 0.2 GeV/c^2; C) 0.5 GeV/c^2; D) 1.0 GeV/c^2
   Note: Correct option: D - 1.0 GeV/c^2

5. **Answer:** A
   Options: A) 24.6 eV; B) 39.5 eV; C) 51.8 eV; D) 54.4 eV
   Note: Correct option: A - 24.6 eV

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'rate of change of the electric flux through S'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '1.5 * 10^8 m/s'.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 25%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** B. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key